Worksheet 4.2 — Graphs of Logarithmic Functions

Grade 12 Advanced Functions (MHF4U) · Unit 4: Exponential and Logarithmic Functions

The Big Ideas

$y = \log_b x$ is the reflection of $y = b^x$ in $y = x$: vertical asymptote $x = 0$, domain $x > 0$, range all reals, through $(1, 0)$. In every step write the **whole function**, e.g. *the argument of* $y = \log(x - 2)$ *must be positive.*

- VA at $x = 0$; domain $x > 0$ (the argument must be positive); range all reals.
- Passes through $(1, 0)$ and $(b, 1)$; increasing for $b > 1$.
- For $y = a \log_b (k(x - d)) + c$ the VA is $x = d$ and points map by $(x, y) \to (\frac{x}{k} + d,\ ay + c)$.

Core Concepts

Inverse of the exponential

$y = \log_b x$ is the reflection of $y = b^x$ in the line $y = x$, with a **vertical** asymptote at $x = 0$.

Domain and range

A logarithmic function is defined only for $x > 0$, and its range is all real numbers the mirror of the exponential.

Transformations

The framework $y = a \log_b(k(x - d)) + c$ shifts and stretches the curve and moves the vertical asymptote to $x = d$.

Writing solutions

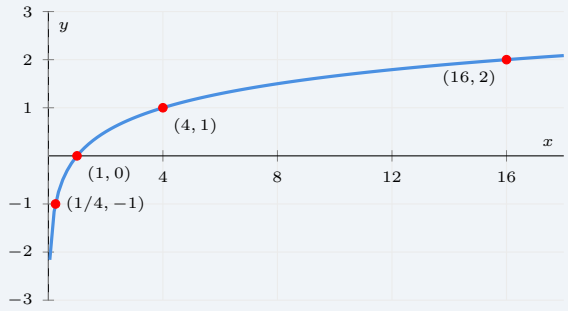
Write the **whole function** in every step. To graph $y = a \log_b(k(x - d)) + c$: draw the asymptote $x = d$, map the parent's key points $\left(\frac{1}{b}, -1\right), (1, 0), (b, 1)$ with $(x, y) \to (\frac{x}{k} + d,\ ay + c)$, then sketch.

Worked Examples

Example 1 — The log graph

Sketch $y = \log_4 x$.

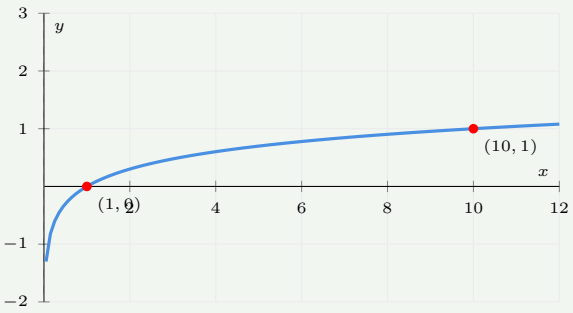
Solution. **Step 1 (vertical asymptote and domain):** The argument of $y = \log_4 x$ must be positive, so the domain is $x > 0$ and the VA is $x = 0$.
Step 2 (key points): From $4^y = x$: $y = -1 \Rightarrow x = \frac{1}{4}$; $y = 0 \Rightarrow x = 1$; $y = 1 \Rightarrow x = 4$; $y = 2 \Rightarrow x = 16$. So $\left(\frac{1}{4}, -1\right), (1, 0), (4, 1), (16, 2)$ are on $y = \log_4 x$.
Step 3 (shape): The base $4 > 1$, so the curve is increasing, with range all reals.
Answer: $y = \log_4 x$: VA $x = 0$, increasing, through $(1, 0)$ and $(4, 1)$.



Example 2 — Domain

Domain of $y = \log x$?

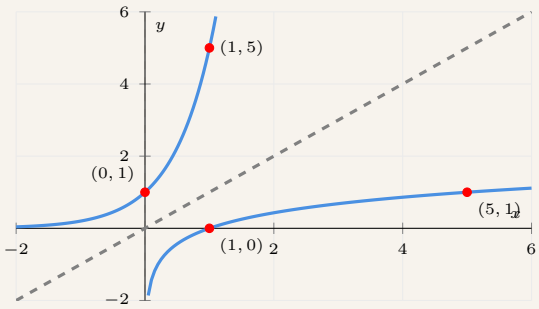
Solution. **Step 1 (positive argument):** The argument of $y = \log x$ must be positive: $x > 0$.
Step 2 (test values): $y = \log 5$ is defined, but $y = \log 0$ and $y = \log(-3)$ are not.
Answer: the domain of $y = \log x$ is $x > 0$.



Example 3 — Inverse pair

How are $y = 5^x$ and $y = \log_5 x$ related?

Solution. **Step 1 (swap x and y):** Start from $y = 5^x$ and swap x and y: $x = 5^y \iff y = \log_5 x$.
Step 2 (points swap): A point (a, b) on $y = 5^x$ becomes (b, a) on $y = \log_5 x$: $(0, 1) \to (1, 0)$ and $(1, 5) \to (5, 1)$.
Step 3 (reflect): The graphs are mirror images across the line $y = x$; the horizontal asymptote $y = 0$ of $y = 5^x$ becomes the vertical asymptote $x = 0$ of $y = \log_5 x$.
Answer: $y = 5^x$ and $y = \log_5 x$ are inverses (reflections in $y = x$).



Example 4 — Vertical asymptote

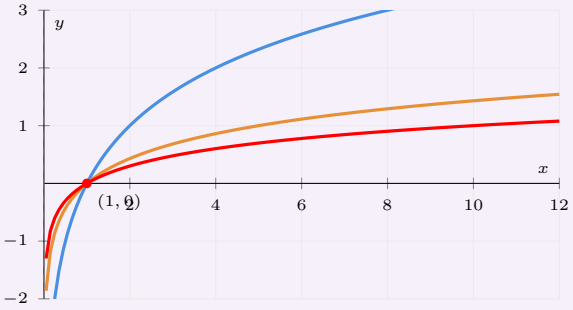
VA of $y = \log x$?

Solution. **Step 1 (look near zero):** For $y = \log x$: $\log 0.1 = -1$, $\log 0.001 = -3$, $\log 0.000001 = -6$. As $x \to 0^+$, $y \to -\infty$.
Step 2 (asymptote): The curve $y = \log x$ falls without bound but never reaches the y-axis.
Answer: the vertical asymptote of $y = \log x$ is $x = 0$.

Example 5 — Key point

What point is on every graph $y = \log_b x$?

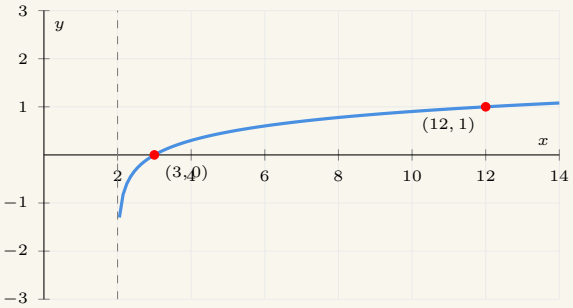
Solution. **Step 1 (log of 1):** For $y = \log_b x$ set $x = 1$: $\log_b 1 = y \iff b^y = 1$, so $y = 0$ for every base b.
Step 2 (also (b, 1)): Since $\log_b b = 1$, the point $(b, 1)$ is also on $y = \log_b x$: $(2, 1)$, $(5, 1)$, $(10, 1)$.
Answer: $(1, 0)$, whatever the base.



Example 6 — Transformation

VA of $y = \log(x-2)$?

Solution. **Step 1 (positive argument):** The argument of $y = \log(x-2)$ must be positive: $x - 2 > 0 \Rightarrow x > 2$.
Step 2 (asymptote): As $x \to 2^+$, $y = \log(x-2) \to -\infty$, so the VA is $x = 2$: the graph of $y = \log x$ has moved right 2.
Step 3 (points): $(1, 0) \to (3, 0)$ and $(10, 1) \to (12, 1)$. Check: $\log(3-2) = \log 1 = 0$ ✓.
Answer: the vertical asymptote of $y = \log(x-2)$ is $x = 2$.



Example 7 — Range

Range of $y = \log x$?

Solution. **Step 1 (can y be any number?):** For $y = \log x$, every real y has an x with $\log x = y$, namely $x = 10^y$.
Step 2 (examples): $y = 5$ comes from $x = 10^5$; $y = -5$ comes from $x = 10^{-5}$.
Answer: the range of $y = \log x$ is all real numbers.

Example 8 — Inverse

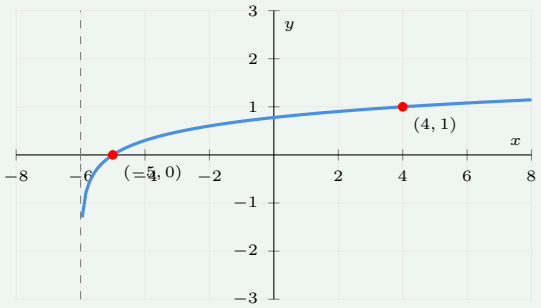
Inverse of $y = 7^x$?

Solution. **Step 1 (swap x and y):** Start from $y = 7^x$ and swap: $x = 7^y$.
Step 2 (logarithmic form): $x = 7^y$ is the same as $y = \log_7 x$.
Step 3 (check with a point): $(1, 7)$ on $y = 7^x$ becomes $(7, 1)$ on $y = \log_7 x$, and $\log_7 7 = 1$ ✓.
Answer: the inverse of $y = 7^x$ is $y = \log_7 x$.

Example 9 — Transformation

VA of $y = \log(x + 6)$?

Solution. **Step 1 (positive argument):** The argument of $y = \log(x + 6)$ must be positive: $x + 6 > 0 \Rightarrow x > -6$.
Step 2 (asymptote): As $x \to -6^+$, $y = \log(x+6) \to -\infty$, so the VA is $x = -6$: the graph of $y = \log x$ has moved left 6.
Step 3 (points): $(1, 0) \to (-5, 0)$ and $(10, 1) \to (4, 1)$. Check: $\log(4 + 6) = \log 10 = 1$ ✓.
Answer: the vertical asymptote of $y = \log(x + 6)$ is $x = -6$.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Domain of $y = \log_6 x$?

Question 2

VA of $y = \log(x + 8)$?

Question 3

What is $\log_b b$?

Question 4

Is $y = \log_7 x$ increasing or decreasing?

Question 5

Inverse of $y = 4^x$?

Question 6

Range of $y = \log x$?

Question 7

VA of $y = \log(x - 5)$?

Question 8

Domain of $y = \log(x - 1)$?

Question 9

Through which point does $y = \log_9 x$ pass on the x-axis?

Question 10

Inverse of $y = 10^x$?

Question 11

VA of $y = \log(x) + 4$?

Question 12

As $x \to 0^+$, what does $y = \log x$ do?

Question 13 — Challenge

State the domain, range, VA and a key point of $y = \log_2(x - 1)$.

Question Recap & Answer Key

Question Recap

1. Domain of $y = \log_6 x$?
2. VA of $y = \log(x + 8)$?
3. What is $\log_b b$?
4. Is $y = \log_7 x$ increasing or decreasing?
5. Inverse of $y = 4^x$?
6. Range of $y = \log x$?
7. VA of $y = \log(x - 5)$?
8. Domain of $y = \log(x - 1)$?
9. Through which point does $y = \log_9 x$ pass on the x-axis?
10. Inverse of $y = 10^x$?
11. VA of $y = \log(x) + 4$?
12. As $x \to 0^+$, what does $y = \log x$ do?
13. State the domain, range, VA and a key point of $y = \log_2(x - 1)$.

✓ Answer Key

1. For $y = \log_6 x$ the argument must be positive, so the domain is $x > 0$
2. For $y = \log(x + 8)$: $x + 8 > 0 \Rightarrow x > -8$, so the VA is $x = -8$
3. $\log_b b = y \iff b^y = b^1$, so $\log_b b = 1$
4. For $y = \log_7 x$ the base $7 > 1$, so the function is increasing
5. Swap x and y in $y = 4^x$: $x = 4^y \iff y = \log_4 x$
6. The range of $y = \log x$ is all reals (e.g. $\log 10^5 = 5$ and $\log 10^{-5} = -5$)
7. For $y = \log(x - 5)$: $x - 5 > 0 \Rightarrow x > 5$, so the VA is $x = 5$
8. For $y = \log(x-1)$: $x-1 > 0 \Rightarrow x > 1$, so the domain is $x > 1$
9. Set $y = 0$ in $y = \log_9 x$: $\log_9 x = 0 \Rightarrow x = 9^0 = 1$, so the point is $(1, 0)$
10. Swap x and y in $y = 10^x$: $x = 10^y \iff y = \log x$ $(= \log_{10} x)$
11. For $y = \log x + 4$ the argument is still x, so $x > 0$ and the VA is $x = 0$ (the $+4$ only moves the graph up)
12. For $y = \log x$: $\log 0.01 = -2$ and $\log 0.0001 = -4$, so $y \to -\infty$
13. For $y = \log_2(x-1)$: $x-1 > 0$ gives domain $x > 1$ and VA $x = 1$; range all reals; key points $(1, 0) \to (2, 0)$ and $(2, 1) \to (3, 1)$, so $(2, 0)$ and $(3, 1)$ are on the graph